Anurag Mishra

Curriculum Vitae
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Research Interests

Mechanistic interpretability of language and time-series foundation models; sparse representation learning; causal circuit analysis and intervention; reasoning and self-correction in large language models; model reliability and evaluation; multimodal detection; and efficient local and edge inference.

Education

Master of Science in Artificial Intelligence 2023 – May 2025
Rochester Institute of Technology *Rochester, NY*

Research focus: mechanistic interpretability, natural language processing, multimodal learning, statistical machine learning, and responsible AI.

Bachelor of Technology in Computer Science and Engineering 2019 – 2023
Sikkim Manipal Institute of Technology *Sikkim, India*

Minor in Artificial Intelligence.

Publications

[1] **Dissecting Chronos: Sparse Autoencoders Reveal Causal Feature Hierarchies in Time Series Foundation Models**
Anurag Mishra
1st Workshop on Time Series in the Age of Large Models (TSALM), **ICLR 2026**. Workshop poster; sole author.
arXiv:2603.10071 · OpenReview · Code

Research contribution. Introduced sparse-autoencoder analysis for Chronos-T5-Large (710M parameters), using 49,152 learned features and 392 causal ablations to reveal a depth-dependent hierarchy across encoder, decoder, and cross-attention representations.

[2] **Automatic Short Answer Grading Using a LSTM Based Approach**
Udit Kr Chakraborty and Anurag Mishra
2023 IEEE World Conference on Applied Intelligence and Computing (AIC), **IEEE, 2023**, pp. 332–337.
doi:10.1109/AIC57670.2023.10263899

Research contribution. Developed an LSTM-based automated-assessment pipeline that models sequential dependencies in variable-length student responses and incorporates semantic similarity into short-answer grading.

[3] **Mechanistic Interpretability of GPT-like Models on Summarization Tasks**
Anurag Mishra
arXiv, **2025**. Sole author.
arXiv:2505.17073 · Code

Research contribution. Localized a summarization-relevant circuit to layers 2, 3, and 5 through differential analysis of GPT-2 internals; targeted LoRA converged in 6 rather than 10 epochs with 75% fewer trainable parameters (31K vs. 124K).

[4] **Advancing Explainability in Neural Machine Translation: Analytical Metrics for Attention and Alignment Consistency**
Anurag Mishra
arXiv, **2024**. Sole author.
arXiv:2412.18669 · Code

Research contribution. Introduced attention-entropy and alignment-agreement metrics for mT5 on an English–German WMT14 subset, showing that sharper attention aligned more closely with external word alignments but did not consistently improve BLEU or METEOR.

[5] **Analyzing the Impact of Climate Change With Major Emphasis on Pollution: A Comparative Study of ML and Statistical Models in Time Series Data**
Anurag Mishra, Ronen Gold, and Sanjeev Vijayakumar
arXiv, **2024**.
arXiv:2405.15835

Research contribution. Compared machine-learning and statistical time-series methods for analyzing and forecasting environmental and pollution-related trends.

Research Experience

Machine Learning Research Assistant, Mechanistic Interpretability Jan 2025 – May 2025
Rochester Institute of Technology *Rochester, NY*

- Independently led an RIT capstone study of how GPT-style models reorganize internal computation during abstractive summarization.
- Built PyTorch/C++ instrumentation for activation tracing, forward hooks, attention analysis, and layer-wise comparison across pre-trained, fully fine-tuned, and LoRA-adapted models.
- Combined KL divergence, attention entropy, and activation-magnitude analyses to localize a task-relevant circuit to layers 2, 3, and 5; 62% of heads in those layers became more focused after fine-tuning.
- Mapped attention heads to topic tracking, coreference resolution, and salience detection, then tested the resulting circuit through targeted parameter-efficient adaptation.
- Presented the work at the RIT Research Symposium, where it received the Best Poster Award.

Machine Learning Research Assistant, LLM Evaluation and Robustness Oct 2024 – Aug 2025
Rochester Institute of Technology, Office of the Provost *Rochester, NY*

- Designed controlled evaluation harnesses for GPT-4, Claude, Llama, and Mistral across multi-step reasoning, planning, factual consistency, world knowledge, and tool-calling reliability.
- Formulated experiments probing causal reasoning, decision-making under uncertainty, adversarial failure modes, and the stability of model behavior across task and prompt variants.
- Developed synthetic benchmarks and retrieval-augmented stress tests to characterize grounding failures and behavioral consistency; the resulting prompting strategy reduced hallucination by 35% on factual-QA benchmarks.
- Built reproducible quality and runtime regression suites and translated experimental findings into technical guidance for responsible institutional deployment.
- The research culminated in a manuscript submitted to Elsevier's *Decision Support Systems*.

Machine Learning Research Assistant, DeFake Project Oct 2024 – Aug 2025
Rochester Institute of Technology, advised by Dr. Matthew Wright *Rochester, NY*

- Developed multimodal CNN–transformer systems for audio-visual lip-sync manipulation detection across 18,000+ videos, improving F1 by 12% over baseline approaches.
- Designed an internal transformer baseline from first principles to establish a controlled reference for testing extensions to the project's state-of-the-art model stack.
- Implemented ring-attention systems from scratch in PyTorch and JAX to study memory-efficient long-sequence training for audio-visual inputs.
- Wrote custom C++/CUDA kernels for CNN–transformer forward and backward paths, improving training efficiency by 40%.
- Generated 50,000+ controlled synthetic samples and broadened adversarial stress testing, improving out-of-distribution generalization by 18%.
- Built Grad-CAM and attention-rollout analyses to inspect cross-modal representations, attention allocation, and failure modes.

Graduate Student Research Assistant Oct 2023 – Sep 2024
National Technical Institute for the Deaf (NTID), Rochester Institute of Technology *Rochester, NY*

- Applied GIS, Python, machine learning, spatial analysis, and cluster detection to emergency-preparedness research serving deaf and hard-of-hearing communities in Rochester and across New York State.
- Analyzed population distribution and geographic access patterns to identify priority areas for emergency planning and service delivery.
- Evaluated New York emergency-management systems and synthesized recommendations for improving accessibility and responsiveness for deaf and hard-of-hearing residents.
- Developed research and course materials on accessible emergency management and collaborated with multidisciplinary teams integrating AI, computer vision, and GIS methods.

Independent Research

Dissecting Chronos: Sparse Autoencoders Reveal Causal Feature Hierarchies in Time Series Foundation Models 2026
Sole-author research project · *ICLR 2026 TSALM Workshop poster* · *Code*

- Built an end-to-end activation-capture and TopK sparse-autoencoder training pipeline for Chronos-T5-Large at six encoder, decoder, and cross-attention extraction points.
- Designed synthetic diagnostics and ETT ablation workflows to connect learned feature semantics to causal forecasting behavior across 49,152 features and 392 single-feature interventions.
- Recovered a depth-dependent hierarchy in which the mid-encoder carried the strongest causal feature while the final encoder represented the broadest semantic taxonomy.

Causal Mechanisms of Backtracking in Reasoning Models 2025
Independent mechanistic-interpretability project · *Code*

- Instrumented DeepSeek-R1-Distill-Qwen-1.5B on GSM8K and used logit-lens analysis to study transitions into revision behavior marked by tokens such as “Wait,” “Actually,” and “Hold on.”
- Tested whether backtracking was mediated by a localized trigger circuit or distributed across the network, using targeted attention- and MLP-layer ablations with random-ablation controls.
- Localized the strongest effect to late MLP layers: ablating MLP layer 27 reduced backtracking by 55%, while ablating layers 19–27 reduced it by 85%; attention-layer ablations increased rather than suppressed the behavior.
- In the phase-one comparison, targeted ablation lowered the backtracking rate from 68.1% to 2.8%, versus 54.9% under random ablation.
- Samples containing backtracking were 2.1× more accurate than samples without it (23.7% vs. 11.5%), linking the localized mechanism to functional performance.

Industry Experience

AI Engineer Aug 2025 – Present
Eaton Ventures Inc. (Rochester Appliances) *Rochester, NY*

- Led internal AI integration across service and business workflows, developing local-first systems and internal tools that turn operational knowledge into structured actions.
- Designed local-inference pipelines for business-internal AI tools and implemented ARM-targeted kernels for model execution in constrained local environments.
- Architected production retrieval and tool-calling services supporting 10,000+ daily requests at 99.9% uptime.
- Reduced end-to-end API latency by 35% through asynchronous orchestration, request-path optimization, timeout and retry handling, and model-serving improvements.
- Established model and system benchmarks spanning quality, latency, throughput, and failure behavior, with regression checks across releases.
- Automated validation and deployment checks in CI/CD workflows, reducing release-cycle time by 60%.

Natural Language Processing Intern Nov 2021 – Feb 2022
Textify AI *Remote*

- Fine-tuned BERT/RoBERTa models for synonym generation and summarization using masked-language-model objectives and domain-specific vocabulary adaptation.
- Prototyped named-entity-recognition and knowledge-graph components to enrich the text-generation and summarization pipeline.
- Built Python/Flask REST services and deployed real-time transformer inference with sub-100ms latency, supporting product integration, testing, and model monitoring.

Machine Learning Intern Jun 2021 – Aug 2021
Defence Research and Development Organisation (DRDO) *New Delhi, India*

- Studied uncertainty-aware decision models using distribution functions, fuzzy-number theory, and Gaussian membership functions; implemented experimental models and authored three technical reports.
- Developed probabilistic sensor-fusion and state-estimation methods for real-time tracking, improving inference speed by 30%.
- Refactored machine-learning prototypes for constrained edge deployment, reducing deployment overhead by 40%.

Honors, Awards, Fellowships & Scholarships

Best Poster Award, RIT Research Symposium 2025
Recognized for research on mechanistic interpretability of GPT-style summarization models.

Second Place, Decentralized Machine Learning Hackathon
Organized by Foundry Digital and FLock.io and hosted by the RIT AI Club; focused on parameter-efficient LLM fine-tuning, LoRA/QLoRA, and model quantization.

RIT Scholarship for the M.S. in Artificial Intelligence 2023
Rochester Institute of Technology.

Prime Minister's Scholarship for Engineering 2019
Awarded through the National Scholarship Portal in support of B.Tech study.

Peer Review Service

Reviewer 2026
Foundation Models for Structured Data (FMSD) Workshop, ICML 2026

Reviewer 2026
Mechanistic Interpretability Workshop, ICML 2026

Reviewer 2026
3rd Workshop on New Trends in AI-Generated Media and Security (AIMS), CVPR 2026

Leadership & Community

Vice President, RIT AI Club
Rochester Institute of Technology

AI/ML Lead, Google Developer Student Club 2021 – 2023
Sikkim Manipal Institute of Technology; mentored 100+ students through introductory AI/ML sessions and technical guidance.

Executive Lead (Debating), Reverbs 2020 – 2022
Sikkim Manipal Institute of Technology; mentored 50+ students in parliamentary and other debate formats.

Technical Skills

Languages	Python, C/C++, CUDA, Rust, SQL
ML frameworks	PyTorch, JAX, TensorFlow, Hugging Face Transformers
Interpretability	Sparse autoencoders, activation tracing, logit lens, attention analysis, circuit analysis, causal interventions, ablation studies
LLM research	Reasoning and tool-calling evaluation, retrieval-augmented generation, supervised fine-tuning, LoRA, synthetic diagnostics
Systems	DDP/FSDP, Docker, Linux, Git, CI/CD, profiling, benchmarking, Weights & Biases, GDB